Imputation

Filling missing values so that methods which cannot cope with them can run, and the price of doing it.

AT A GLANCE

Where	Data → Imputation
Data levels	Protein and phosphosite. Phosphoproteins cannot be imputed here at all.
Needs	An annotated dataset. The matrix imputed is the filtered one if filtering has been run, otherwise the annotated $\log_2$ matrix.
Two methods	**Quantile**, the default, draws around a low quantile of the observed values, in the spirit of MinProb (Lazar et al. 2016); **Numerical** is the Perseus downshift (Tyanova et al. 2016), drawing around the mean shifted down by a multiple of the standard deviation.
Two scopes	Sample-wise, one draw distribution per sample, or global, one for the whole matrix.
Stores	Imputed Data, an additional matrix a panel with a **Data Source** selector can be pointed at.
Reproducible	Keep the seed, input, order, parameters and software environment; the report does not state the seed. For exact replay, retain the realised imputed matrix.

1 What the page answers

Imputation supplies model-based values, not measurements. It fills NA in eligible sample columns with a value drawn from a distribution placed where an unobserved measurement plausibly would be — low, narrow, below the bulk of the observed data — so that a method which cannot accept a gap can be run at all. Nothing about a drawn value is a measurement.

The page answers two questions: what the drawn values will look like beside the values that were measured, which is what the three figures show, and how much of the matrix they will be, which is what the cards under **Applied imputation** report. A matrix that is two per cent

imputed and one that is forty per cent imputed are different experiments, and the second is mostly the draw distribution. The figures are a preview — redrawn from the controls, touching nothing — and only **Impute Data Values** stores a matrix.

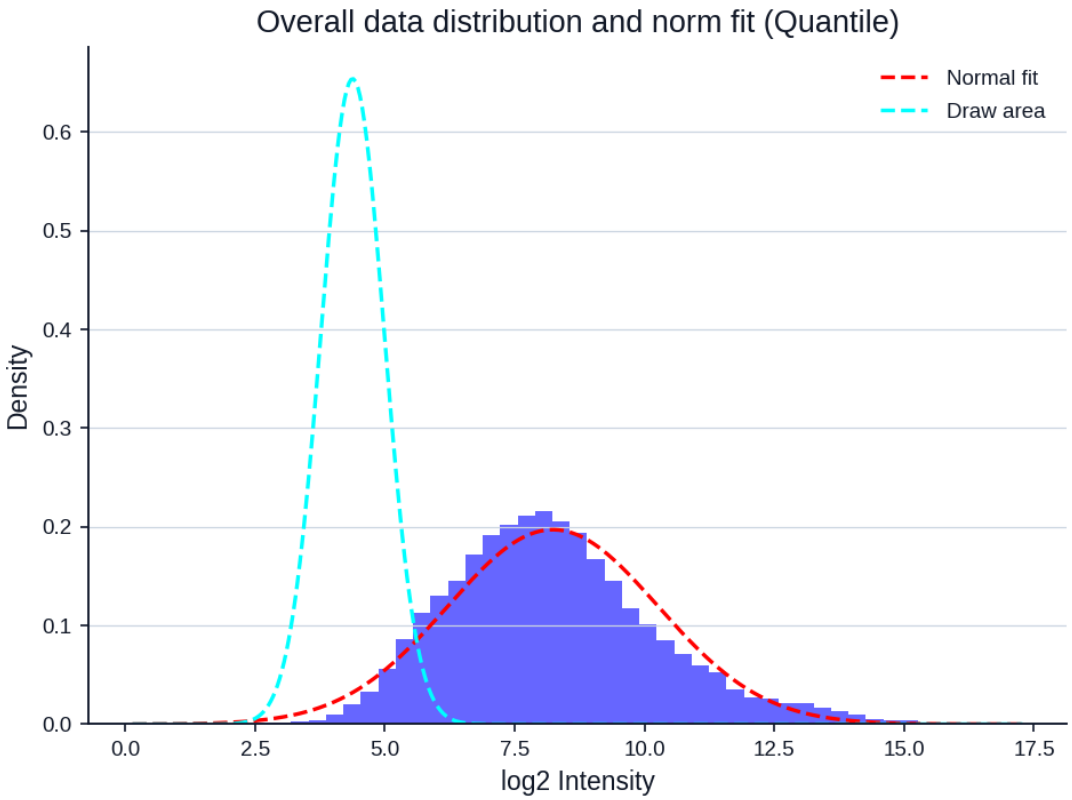


Figure 1.1. The observed distribution with its normal fit and the draw area, on the example data at protein level with the quantile method: every observed $\log_2$ protein intensity as a density, peaking near 8 and running from roughly 2.5 to 15, with the broad dashed normal fit over it and the narrow dashed draw area centred near 4.5, well below the bulk. The legend names the two curves and the title names the method.

Reading it

The figure above is the second of the three the page draws. The first is not reproduced here: it compares rows with no missing value against rows with at least one, and says whether missingness is concentrated in the low-intensity rows, which is the assumption the whole draw model rests on. The third is the after plot, which is the subject of the last bullet below.

- Every observed value in the sample columns, pooled over all samples, as a density.
- The red curve: a normal density with the mean and the standard deviation of those values. It is fitted to the whole distribution rather than its lower tail, and says how far from normal the data are before anything is drawn.
- The cyan curve: the draw area, the density the missing values are taken from. Sample-wise it is the mixture of the per-sample draw distributions, each weighted by the number of values that sample contributes; globally it is the one distribution.

- After the draw: measured values and invented values as two histograms on common bins. The imputed set should sit under the left shoulder of the observed one rather than displacing it.

> **EASY TO GET WRONG**
>
> Every plot drawn from Imputed Data will look better than the experiment was: no gaps are left in it, so a heatmap has no white, a correlation is computed on complete columns, and a coverage count would reach the number of rows. Impute only for the method that requires it and keep the non-imputed matrix for anything whose subject is completeness — which is why coverage and the missing-value pages never read the imputed matrix at all.

2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Protein or phospho. Phosphoprotein is not offered.	Protein	page
Method	Quantile or Numerical: where the draw is centred, and how its width is set.	Quantile	page
Sample-wise imputation	On: every sample gets its own draw distribution, from its own observed values. Off: one distribution for the whole matrix.	on	page
Quantile	Quantile method only. The quantile of the observed values the draw is centred on; 0.01 is the first percentile.	0.01	page
Adjust standard deviation	Quantile method only. Draw width as a multiple of the standard deviation of the observed values. Must be greater than 0; 0 is refused.	0.3	page
Downshift	Numerical method only. How many standard deviations below the mean the draw is centred.	1.8	page
Adjust standard deviation	Numerical method only, and a different parameter from the field of the same name above: draw width as a multiple of the standard deviation. Must be greater than 0; 0 is refused.	0.3	page
Random seed	Fixes the draw, so the same parameters reproduce the matrix value for value.	1337	page

Controls on this page (continued)

Control	What it does	Default	Where
Impute Data Values	Computes the imputation and stores it as this level's Imputed Data. Nothing else on the page stores anything.	—	page
Show imputation draw distribution	Draws the cyan draw-area curve over the middle figure.	on	page
Separately / Together	How the two histograms of the lower figure are scaled: each to its own density, or both by the combined count.	Separately	page

3 How it is computed

WHICH MATRIX IS IMPUTED

There is no **Data Source** selector here: the most processed matrix available is taken — the filtered matrix if filtering has been run, otherwise the annotated $\log_2$ matrix, otherwise the uploaded frame, which is reported as a warning. A filter that kept no rows stops the run with an error rather than quietly imputing the unfiltered matrix. Only NA counts as missing. Zero on the $\log_2$ scale is an observed value, because $2^0 = 1$, and is never drawn over; the convention that a raw intensity of zero means *not detected* is applied earlier.

THE DRAW

Let x be the observed values in scope — one sample's column when sample-wise, the whole matrix when global — with mean $\bar{x}$ and standard deviation s, taken with $n - 1$ in the denominator. Each missing value in that scope is replaced by an independent draw from

$$\mathcal{N}\left(Q_q(x),\, (a\,s)^2\right) \quad \text{or} \quad \mathcal{N}\left(\bar{x} - d\,s,\, (w\,s)^2\right),$$

the first for **Quantile**, the second for **Numerical**. Q_q is the q-quantile and q is **Quantile**; a, d and w are **Adjust standard deviation**, **Downshift** and the numerical **Adjust standard deviation**. At the defaults the quantile draw is centred on the $q = 0.01$ quantile of the observed values with a spread of $0.3\,s$ — a MinProb-like draw (Lazar et al. 2016) — and the numerical draw, the Perseus downshift (Tyanova et al. 2016),

1.8 s below their mean with the same spread. Both spreads must be greater than 0: a spread of 0 would give every imputed value in a sample the same number, and such constant replicates make a later t-test undefined, so the request is refused with Adjust standard deviation must be greater than 0.
Sample-wise, $\bar{x}$, s and $Q_q(x)$ are recomputed inside every sample column, so a sample that ran low is filled low and its imputed values stay in the company of its own measurements. Globally, one centre and one width are computed from all observed values pooled; this needs at least two observed values with non-zero spread and errors otherwise. The generator is created per request and seeded with **Random seed**.

Edge cases

- Sample-wise, a sample with fewer than two observed values or no spread among them has nothing to estimate a distribution from. It is skipped, its missing values survive the run, and a warning under the plots names it.
- Number imputed can therefore be smaller than the number of missing values the matrix started with: the cards report what was done, not what was asked.
- A phosphoprotein level is refused outright. Impute the phosphosites, then collapse from the imputed phosphosite source.

4 What it changes downstream

Imputation adds a matrix and replaces nothing: Data and Filtered Data remain as they were.

- Panels with a **Data Source** selector can be pointed at Imputed Data: PCA, the statistics pages, the comparison pipeline, Kinase Library, the single-protein page and the analysis canvas. One opened for the first time after a successful run opens on the most processed matrix that exists, which is now the imputed one; one that already holds a choice keeps it.
- Coverage, the missing-value plots and tables, the histograms, the boxplots, the correlation surfaces and the distribution page offer no such choice and do not follow the imputed matrix: they describe the data as annotated, which is the only way to go on answering *what was measured*.
- Every statistic computed on the imputed matrix counts drawn values as data: a test across a condition in which a feature was never measured compares two draw distributions. Check such a feature on the missing-value heatmap before believing it.

5 Normalization

Normalization is not on this page — it is the **Normalization** control on Data → Annotation, applied when the annotation is generated — but it is the other step that rewrites the values, and it runs before both filtering and imputation. Its values are None (the default), Median centering, Quantile normalization and VSN, and which one ran is stated in the annotation output. Left at None, sample loading and instrument drift are uncorrected unless the upload was already normalized upstream — MaxLFQ or PG.Normalised, say.

WHAT EACH ONE DOES

All three are global, over all sample columns at once; normalising within conditions would remove the differences the later tests look for.
Median centering works on the $\log_2$ scale, where a multiplicative loading difference is an additive offset: each sample's median is subtracted and the grand median of those medians added back. The sample-specific offsets can change fold changes; the shared constant added afterwards cancels in a comparison.
Quantile normalization sorts each sample's observed values, interpolates them onto a common grid with as many positions as the matrix has rows, and averages those curves into one reference distribution, off which every sample is read at its own quantile positions. Missing values neither are filled nor move a sample's ranks.
VSN is fitted on linear intensities and returns a generalised-log scale approaching $\log_2$ for large intensities, so it replaces the $\log_2$ transform rather than following it. NA is carried through.

VSN is delegated to the Bioconductor vsn package rather than reimplemented, so the option appears in the list only where Rscript is present *and* that package is installed in it; otherwise the list has three entries, and where R is there but the package is not, the message names the installation command. The annotation output records the package version that ran. Phosphoprotein data is never normalized at the phosphoprotein level, and the **Normalization** control is not shown there.

6 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
imputed = impute_values(data, meta)
fig = impute_values(data, meta, ret=2)
```

```r
imputed <- impute_values(data, meta)
fig <- impute_values(data, meta, ret = 2)
```

One function does both jobs, and ret chooses which: 0 returns the imputed matrix, 1 the missing-value comparison, 2 the figure above and 3 the after plot, so the page's three figures and its stored matrix all come out of the same call. Every other argument is a control on the page under the same name and at the same default — q, adj_std, sample_wise, seed, method, down_shift and width. A fixed seed supports repeatability within the same implementation, environment and input order; Python and R need not generate identical draws. The one thing missing from ret=2 is the cyan draw-area curve: the package plots the observed histogram and the normal fit, and the application adds the draw area over them.

7 Exporting

The diagnostic figures intentionally have no plot or notebook download. **Download Full Imputed CSV** writes the stored matrix as imputed_protein.csv or imputed_phospho.csv, using the parameters of the applied run rather than the ones currently in the fields, so what is downloaded is what the rest of the session is using. The table at the foot of the page is its first twenty rows.

8 Further reading

The vignette *Missing value handling* of the Bioconductor package DEP (Bioconductor 3.22) provides worked proteomics examples of filtering and different imputation assumptions. It is useful methodological background, not a claim that CoPYlot implements every DEP option or reproduces its random draws.

Behind this page

Function	What it produces
impute_values	The draw model in both packages: same parameters, aligned method and parameters; Python and R random draws need not be identical
impute_values_with_diagnostics	The same draw, returning also the before and after value sets, the draw components and any skipped samples
run_imputation	Runs it and stores the session's imputed matrix
imputed_dataframe	The imputed matrix alone, for the CSV download
normalize_data	Median, quantile and VSN normalization of a $\log_2$ matrix
vsn_normalize_linear	The call into Bioconductor vsn